Elementary Analysis

Chapter 2, Section 15

Alternating Series and Integral Tests
Exercises 15.1–15.8, with every subpart

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

15 Alternating series and integral tests **2**

15.1 Exercise 15.1: Two alternating series with very different magnitudes 2
15.2 Exercise 15.2: Powers of periodically sampled sine values 3
15.3 Exercise 15.3: The logarithmic refinement of the harmonic series 4
15.4 Exercise 15.4: Four series involving logarithms 5
15.5 Exercise 15.5: Why a direct comparison does not establish all positive powers . . . 6
15.6 Exercise 15.6: What squaring the terms can change 7
15.7 Exercise 15.7: A necessary decay rate for decreasing summable terms 8
15.8 Exercise 15.8: A general integral test and its error bounds 9

15 Alternating series and integral tests

In this section, log means the natural logarithm. A series is *conditionally convergent* if it converges but its series of absolute values diverges. Changing or removing finitely many terms does not change convergence, by Exercise 14.9. We use this fact when a comparison or monotonicity estimate holds only after a fixed index.

The alternating-series theorem requires nonnegative magnitudes that decrease to zero. An alternating sign by itself does not guarantee convergence. An integral test requires a nonnegative decreasing function matching the terms; the precise statement and proof are given in Exercise 15.8 below. As in the textbook's Section 15, the usual elementary integral and derivative formulas are available.

15.1 Exercise 15.1: Two alternating series with very different magnitudes

What we must show. Decide whether the series in parts (a) and (b) converge, and justify each answer.

Plan. Check the magnitudes before applying the alternating-series theorem. In particular, a convergent series must have terms tending to zero, regardless of their signs.

(a) The series $\sum_{n=1}^{\infty}(-1)^n/n$.

Put $b_n = 1/n$. For every positive integer n, we have $b_n > 0$. Also $n+1 > n > 0$, so taking reciprocals reverses the inequality:

$$b_{n+1} = \frac{1}{n+1} < \frac{1}{n} = b_n.$$

Finally, $b_n \to 0$. Indeed, for $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. If $n > N$, then $|b_n - 0| = 1/n < \varepsilon$.

Thus $\sum_{n=1}^{\infty}(-1)^{n+1}b_n$ converges by the alternating-series theorem. Multiplication by -1 preserves convergence, by Exercise 14.5(b), and gives the series requested here. Its series of absolute values is $\sum_{n=1}^{\infty} 1/n$, the divergent harmonic series. Hence the requested series converges conditionally.

(b) The series $\sum_{n=1}^{\infty}(-1)^n n!/2^n$.

Let $c_n = n!/2^n > 0$. Cancellation of the common factorial and power gives

$$\frac{c_{n+1}}{c_n} = \frac{(n+1)!}{2^{n+1}}\frac{2^n}{n!} = \frac{n+1}{2}.$$

For $n \geq 3$, this ratio is at least 2. Repeated multiplication therefore gives, for $n \geq 3$,

$$c_n \geq c_3\, 2^{n-3} = \frac{3}{4}\, 2^{n-3}.$$

The right side tends to $+\infty$. In particular c_n does not tend to zero. If the signed terms $(-1)^n c_n$ tended to zero, their absolute values c_n would tend to zero too, a contradiction. The necessary term condition, Corollary 14.5, now proves divergence. Alternation does not compensate for these growing magnitudes.

Conclusion. Part (a) converges conditionally. Part (b) diverges because its terms do not tend to zero.

15.2 Exercise 15.2: Powers of periodically sampled sine values

What we must show. Determine convergence of

$$\text{(a)} \quad \sum_{n=1}^{\infty} \left[\sin\left(\frac{n\pi}{6}\right)\right]^n, \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \left[\sin\left(\frac{n\pi}{7}\right)\right]^n.$$

The exponent n applies to the whole sine value in each case.

Plan. In part (a), the sampled sine values reach 1 infinitely often. In part (b), there are only finitely many sampled values, and every one has absolute value strictly less than 1.

(a) Exhibit terms that stay equal to 1.

For every integer $k \geq 0$, choose $n_k = 12k + 3$. These are strictly increasing positive indices, and

$$\frac{n_k\pi}{6} = 2k\pi + \frac{\pi}{2}.$$

Sine has period 2π and $\sin(\pi/2) = 1$, so

$$\left[\sin\left(\frac{n_k\pi}{6}\right)\right]^{n_k} = 1^{n_k} = 1.$$

The terms of a convergent series must tend to zero, and every subsequence of a sequence tending to zero must also tend to zero. This constant subsequence contradicts that requirement. Therefore the series in part (a) diverges.

(b) Bound all the absolute sine values by one number below 1.

The values $\sin(n\pi/7)$ repeat when n is increased by 14, because the angle then increases by 2π. Thus all their absolute values occur in the finite list

$$\left\{\left|\sin\left(\frac{r\pi}{7}\right)\right| : r = 1, 2, \ldots, 14\right\}.$$

None of these absolute values equals 1. Otherwise the angle $r\pi/7$ would have the form $\pi/2 + \ell\pi$ for some integer ℓ. After division by π and multiplication by 14, this would give $2r = 7(2\ell + 1)$. The left side is even and the right side is odd, which is impossible.

Let c be the maximum of that finite list. Every entry is less than 1, so its attained maximum satisfies $c < 1$ as well; also $c \geq 0$. In fact $c > 0$, since $\sin(\pi/7) > 0$. For every $n \geq 1$,

$$\left|\left[\sin\left(\frac{n\pi}{7}\right)\right]^n\right| = \left|\sin\left(\frac{n\pi}{7}\right)\right|^n \leq c^n.$$

The inequality is preserved when taking the positive integer power because both sides are nonnegative. The geometric series $\sum_{n=1}^{\infty} c^n$ converges, since $0 < c < 1$. Comparison proves that the series of absolute values converges. Hence the series in part (b) converges absolutely.

Conclusion. Part (a) diverges by the necessary term condition. Part (b) converges absolutely by comparison with a geometric series.

15.3 Exercise 15.3: The logarithmic refinement of the harmonic series

What we must show. For a real parameter p, prove

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p} \text{ converges} \quad \Longleftrightarrow \quad p > 1.$$

Plan. For positive p, use the integral test with the substitution $u = \log x$. For nonpositive p, compare directly with $1/n$.

1. Check positivity and the case $p \leq 0$.

For $n \geq 2$, $\log n > 0$, so every real power $(\log n)^p$ is defined and positive. Choose an integer $N > e$. For $n \geq N$, $\log n > 1$. If $p \leq 0$, then $(\log n)^p \leq 1$, and therefore

$$\frac{1}{n(\log n)^p} \geq \frac{1}{n}.$$

The harmonic series diverges. Comparison shows that the series under consideration diverges to $+\infty$ for $p \leq 0$.

2. For $p > 0$, verify the hypotheses of the integral test.

Define

$$f_p(x) = \frac{1}{x(\log x)^p} \qquad (x \geq 2).$$

The function is positive and continuous on this interval. Differentiating the product $x^{-1}(\log x)^{-p}$ gives

$$\begin{aligned} f_p'(x) &= -x^{-2}(\log x)^{-p} - px^{-2}(\log x)^{-p-1} \\ &= -\frac{\log x + p}{x^2(\log x)^{p+1}} < 0. \end{aligned}$$

The denominator is positive, and the numerator $\log x + p$ is positive because $x \geq 2$ and $p > 0$. Thus the function decreases and matches the series terms at every integer $n \geq 2$.

3. Evaluate the integrals and distinguish the parameter cases.

For $R \geq 2$, put $u = \log x$, so $du = dx/x$. Then

$$\int_2^R \frac{dx}{x(\log x)^p} = \int_{\log 2}^{\log R} u^{-p}\, du.$$

If $p \neq 1$, the power rule gives

$$\int_2^R \frac{dx}{x(\log x)^p} = \frac{(\log R)^{1-p} - (\log 2)^{1-p}}{1-p}.$$

For $p > 1$, one has $1 - p < 0$, so $(\log R)^{1-p} \to 0$ as $R \to \infty$. The integral tends to the finite positive number $(\log 2)^{1-p}/(p-1)$. The integral test proves convergence.

For $0 < p < 1$, one has $1 - p > 0$, so $(\log R)^{1-p} \to +\infty$. The denominator $1 - p$ is positive, and the integral tends to $+\infty$. The integral test proves divergence.

For $p = 1$, integration instead gives

$$\int_2^R \frac{dx}{x \log x} = \log(\log R) - \log(\log 2) \longrightarrow +\infty.$$

The arguments of both outer logarithms are positive, even though $\log(\log 2)$ itself is negative. This case also diverges.

Conclusion. Exactly the parameters $p > 1$ give convergence. For every $p \leq 1$, the nonnegative series diverges to $+\infty$.

15.4 Exercise 15.4: Four series involving logarithms

What we must show. Determine convergence of each of the four requested series.

Plan. Compare the first two series with the harmonic series. For the other two, check the integral-test hypotheses and compute the relevant integrals.

(a) The series $\sum_{n=2}^{\infty} 1/(\sqrt{n} \log n)$.

We first justify the eventual inequality $\log x \leq \sqrt{x}$. Let $g(x) = \sqrt{x} - \log x$ for $x > 0$. Its derivative is

$$g'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{x} = \frac{\sqrt{x} - 2}{2x} \geq 0 \qquad (x \geq 4).$$

Also $\log 2 = \int_1^2 dt/t < 1$, since $1/t < 1$ on $(1, 2]$. Thus $g(4) = 2 - \log 4 = 2 - 2\log 2 > 0$. As g is nondecreasing on $[4, \infty)$, we have $\log x < \sqrt{x}$ there.

For integers $n \geq 4$, both logarithms and square roots are positive, and the inequality just proved gives

$$\sqrt{n} \log n \leq n, \qquad \frac{1}{\sqrt{n} \log n} \geq \frac{1}{n}.$$

Taking reciprocals reverses the positive-denominator inequality. Comparison with the divergent harmonic series proves divergence to $+\infty$.

(b) The series $\sum_{n=2}^{\infty} (\log n)/n$.

Choose an integer $N > e$. If $n \geq N$, monotonicity of the logarithm gives $\log n > \log e = 1$. Consequently

$$\frac{\log n}{n} \geq \frac{1}{n}.$$

All terms are positive for $n \geq 2$. Comparison with the harmonic series, with the first finitely many indices ignored, proves divergence to $+\infty$.

(c) The series $\sum_{n=4}^{\infty} 1/[n(\log n)(\log \log n)]$.

For $x \geq 4$, we have $\log x > 1$. One direct check at the left endpoint is

$$\log 2 = \int_1^2 \frac{dt}{t} > \frac{1}{2}, \qquad \log 4 = 2 \log 2 > 1.$$

Therefore $\log \log x > 0$ on this whole interval. The three factors x, $\log x$, and $\log \log x$ are positive, continuous, and increasing. Their product is increasing, so its positive reciprocal

$$f(x) = \frac{1}{x(\log x)(\log \log x)}$$

is decreasing and continuous on $[4, \infty)$.

Two substitutions, first $u = \log x$ and then $v = \log u$, give

$$\begin{aligned}\int_4^R \frac{dx}{x(\log x)(\log\log x)} &= \int_{\log 4}^{\log R} \frac{du}{u \log u} \\ &= \int_{\log\log 4}^{\log\log R} \frac{dv}{v} \\ &= \log(\log(\log R)) - \log(\log(\log 4)).\end{aligned}$$

Every argument of a logarithm in these formulas is positive for $R \geq 4$: in particular $\log\log 4 > 0$. As $R \to \infty$, the first triple logarithm tends to $+\infty$. The integral test proves that this series diverges to $+\infty$.

(d) The series $\sum_{n=2}^{\infty} (\log n)/n^2$.

Let $h(x) = (\log x)/x^2$ for $x \geq 2$. This function is positive and continuous. Differentiation gives

$$h'(x) = \frac{1 - 2\log x}{x^3} < 0 \qquad (x \geq 2),$$

because $\log x \geq \log 2 > 1/2$, as established in part (c). Thus the integral test applies. For integration by parts, take $u = \log x$ and $dv = x^{-2}\,dx$. Then $du = dx/x$ and $v = -1/x$, so

$$\begin{aligned}\int_2^R \frac{\log x}{x^2}\,dx &= \left[-\frac{\log x}{x}\right]_2^R + \int_2^R \frac{dx}{x^2} \\ &= \frac{\log 2 + 1}{2} - \frac{\log R + 1}{R}.\end{aligned}$$

Part (a) gives $0 \leq (\log R)/R \leq 1/\sqrt{R}$ for $R \geq 4$. The right side tends to zero, as does $1/R$. Hence the integral tends to $(\log 2 + 1)/2$, a finite number. The integral test proves convergence of the series.

Conclusion. Parts (a), (b), and (c) diverge to $+\infty$. Part (d) converges.

15.5 Exercise 15.5: Why a direct comparison does not establish all positive powers

What we must show. Explain why the text used an integral argument to prove the convergence of $\sum 1/n^p$ for every $p > 1$.

Plan. A comparison proof needs an independently known convergent upper series. It cannot assume a nearby case of the very theorem it is trying to establish.

1. Check which direction of comparison is useful.

Knowing that $\sum 1/n^2$ converges immediately handles $p \geq 2$, since $0 \leq 1/n^p \leq 1/n^2$ for $n \geq 1$. For $1 < p < 2$, however, the inequality reverses:

$$\frac{1}{n^p} \geq \frac{1}{n^2}.$$

Being larger than the terms of a convergent nonnegative series does not establish either convergence or divergence. Comparison with $1/n$ also fails to settle the question: for $p > 1$ the terms $1/n^p$ are smaller than the terms of that divergent series.

2. Explain the missing starting exponent.

If convergence were already known at some $p_0 > 1$, comparison would prove it for every $p \geq p_0$. It would leave the interval $1 < p < p_0$ unresolved. There is no smallest real exponent greater than 1: between 1 and p_0 lies $(1 + p_0)/2$. Thus no single known case $p_0 > 1$ settles the whole range by this direct comparison. Choosing an unproved exponent q with $1 < q < p$ merely replaces the original convergence problem with another case of the same problem. The integral calculation supplies an independent argument for every $p > 1$.

3. Comparison can still work after grouping the terms.

The explanation above does not mean a comparison proof is impossible. For example, fix $p > 1$ and group the indices into blocks

$$2^j \leq n < 2^{j+1} \qquad (j = 0, 1, 2, \ldots).$$

There are exactly 2^j integers in this block, and each satisfies $n^{-p} \leq 2^{-jp}$. Hence the sum of the block is at most

$$2^j 2^{-jp} = 2^{j(1-p)}.$$

The bounding block sums form a geometric series with ratio $2^{1-p} < 1$. Every original partial sum is bounded above by the sum of this geometric series, since its terms are nonnegative and its indices occupy finitely many blocks. The original partial sums are increasing and bounded above, so they converge by Theorem 10.2. This is a valid alternative proof using an extra grouping argument.

Conclusion. The integral proof avoids relying on an unproved nearby p-series. A plain comparison with one known exponent does not cover all $p > 1$; a geometric comparison of grouped blocks does.

15.6 Exercise 15.6: What squaring the terms can change

What we must show. Supply the examples in parts (a) and (c), and justify the nonnegative-term assertion in part (b).

(a) A divergent series whose squared-term series converges.

Take $a_n = 1/n$. Then $\sum a_n$ is the divergent harmonic series, whereas

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges by the p-series theorem, with $p = 2 > 1$. These two facts verify both required properties of the example.

(b) Squaring preserves convergence when the original terms are nonnegative.

Suppose $a_n \geq 0$ and $\sum a_n$ converges. Corollary 14.5 gives $a_n \to 0$. There is therefore an integer N with $0 \leq a_n < 1$ whenever $n > N$. Multiplication of $a_n < 1$ by the nonnegative number a_n gives

$$0 \leq a_n^2 \leq a_n \qquad (n > N).$$

The comparison test proves convergence of $\sum_{n>N} a_n^2$. Adding its finitely many omitted real terms preserves convergence, so the whole squared-term series converges. This is also Exercise 14.7 with the power $p = 2$.

(c) A convergent signed series whose squared-term series diverges.

Take

$$a_n = \frac{(-1)^{n+1}}{\sqrt{n}}.$$

Its magnitudes $b_n = 1/\sqrt{n}$ are positive and decrease, because the positive square root increases with its argument and taking reciprocals reverses order. They tend to zero: given $\varepsilon > 0$, choose $N > 1/\varepsilon^2$; then $n > N$ implies $1/\sqrt{n} < \varepsilon$. The alternating-series theorem proves that $\sum a_n$ converges. However,

$$a_n^2 = \frac{[(-1)^{n+1}]^2}{(\sqrt{n})^2} = \frac{1}{n},$$

so $\sum a_n^2$ is the divergent harmonic series.

Conclusion. Use $a_n = 1/n$ for part (a) and $a_n = (-1)^{n+1}/\sqrt{n}$ for part (c). Part (b) follows from the eventual bound $0 \le a_n^2 \le a_n$.

15.7 Exercise 15.7: A necessary decay rate for decreasing summable terms

What we must show. Prove that if (a_n) is a decreasing real sequence and $\sum a_n$ converges, then $na_n \to 0$. Then apply this conclusion to the harmonic series.

Plan. First deduce that every a_n is nonnegative. A late block of roughly $n/2$ terms is small by the Cauchy criterion, yet every term in that block is at least a_n.

(a), first: derive the sign condition from the actual hypotheses.

Since $\sum a_n$ converges, Corollary 14.5 gives $a_n \to 0$. The word decreasing means $a_{n+1} \le a_n$; strict decrease is not required. If $a_j < 0$ at any index j, then $a_n \le a_j < 0$ for all $n \ge j$. Such a tail cannot tend to zero: with $\varepsilon = -a_j/2 > 0$, its terms satisfy $|a_n| \ge -a_j > \varepsilon$. This contradicts $a_n \to 0$. Thus $a_n \ge 0$ for every index n.

(a), second: use a tail block with an explicit number of terms.

Fix $\varepsilon > 0$. By the Cauchy criterion for the convergent series, choose an integer $N \ge 1$ such that

$$q \ge m > N \quad \Longrightarrow \quad \left| \sum_{j=m}^{q} a_j \right| < \frac{\varepsilon}{2}.$$

For any integer $n > 2N$, set $k = \lfloor n/2 \rfloor$. Here $\lfloor x \rfloor$ is the greatest integer at most x. We have $k \ge N$, so the block starting at $k+1$ begins after N. It contains exactly $n - k$ terms, and $n - k \ge n/2$. For $k + 1 \le j \le n$, decreasingness gives $a_j \ge a_n$. Consequently

$$0 \le \frac{n}{2} a_n \le (n-k) a_n \le \sum_{j=k+1}^{n} a_j = \left| \sum_{j=k+1}^{n} a_j \right| < \frac{\varepsilon}{2}.$$

The equality with the absolute value uses nonnegativity of the terms. Multiplying the resulting strict upper bound by 2 gives $0 \le na_n < \varepsilon$ whenever $n > 2N$. This verifies the definition of $na_n \to 0$.

(b) Apply the necessary condition to $a_n = 1/n$.

The sequence $1/n$ is decreasing. If the harmonic series converged, part (a) would imply

$$n\left(\frac{1}{n}\right) \longrightarrow 0.$$

The expression on the left equals 1 for every n, so its limit is 1, a contradiction. Therefore the harmonic series diverges. Since its terms are nonnegative, its partial sums are increasing; having no finite limit, they tend to $+\infty$ by the monotone sequence theorem for unbounded increasing sequences.

Conclusion. Every decreasing real sequence with a convergent series satisfies $na_n \to 0$. The constant value $n(1/n) = 1$ therefore provides another proof of divergence of the harmonic series.

15.8 Exercise 15.8: A general integral test and its error bounds

What we must show. State and prove a precise integral test of the type described in Discussion 15.2.

Plan. Compare each unit-width rectangle with the integral below and above it. Nonnegativity makes both the partial sums and the integrals increasing, so a finite upper bound implies convergence.

1. State the assumptions and the conclusion.

Let $m \geq 1$ be an integer, and let $f : [m, \infty) \to [0, \infty)$ be continuous and nonincreasing. Set $a_n = f(n)$ for every integer $n \geq m$. Define the improper integral by

$$\int_m^\infty f(x)\,dx = \lim_{R\to\infty} \int_m^R f(x)\,dx,$$

where the value $+\infty$ is allowed. Then

$$\sum_{n=m}^{\infty} a_n \text{ converges} \quad \Longleftrightarrow \quad \int_m^\infty f(x)\,dx < +\infty.$$

If either side is infinite, both the integral and the series diverge to $+\infty$. If they converge and $R_N = \sum_{n=N+1}^{\infty} a_n$ for $N \geq m$, then additionally

$$\int_{N+1}^\infty f(x)\,dx \leq R_N \leq \int_N^\infty f(x)\,dx.$$

Continuity is a sufficient condition ensuring that all the finite integrals used here exist. The proof also works whenever those integrals exist, with the same nonnegativity and monotonicity assumptions.

2. Establish the unit-interval inequalities.

Fix an integer $k \geq m$. If $k \leq x \leq k+1$, nonincreasingness gives $f(k+1) \leq f(x) \leq f(k)$. Integrating these constant bounds over the interval of length 1 yields

$$f(k+1) \leq \int_k^{k+1} f(x)\,dx \leq f(k).$$

In particular, summing the right-hand inequalities for $k = m, m+1, \ldots, N$ gives

$$\int_m^{N+1} f(x)\,dx \leq \sum_{k=m}^{N} f(k).$$

For the opposite comparison, use $f(k) \le \int_{k-1}^{k} f(x)\,dx$ for $k \ge m+1$. Keep the first term $f(m)$ separate and sum to obtain

$$\sum_{k=m}^{N} f(k) \le f(m) + \int_{m}^{N} f(x)\,dx.$$

At $N = m$, the second sum has no remaining terms and the integral is zero, so this estimate still holds. Combining the two bounds, with $S_N = \sum_{k=m}^{N} f(k)$, gives

$$\boxed{\int_{m}^{N+1} f(x)\,dx \le S_N \le f(m) + \int_{m}^{N} f(x)\,dx.}$$

3. A finite integral bounds the partial sums.

Suppose the improper integral has finite value I. Since $f \ge 0$, one has $\int_m^N f \le I$. Therefore $0 \le S_N \le f(m) + I$ for every $N \ge m$. Also $S_{N+1} - S_N = f(N+1) \ge 0$, so (S_N) is increasing. The monotone sequence theorem gives a finite limit of (S_N), which is precisely convergence of the series.

4. A convergent series bounds all the finite integrals.

Conversely, suppose $S_N \to S < +\infty$. The nonnegative terms imply $S_N \le S$. If $R \ge m$ is any real number, choose an integer $N \ge m$ with $R \le N+1$. Nonnegativity and the first rectangle estimate give

$$0 \le \int_{m}^{R} f(x)\,dx \le \int_{m}^{N+1} f(x)\,dx \le S_N \le S.$$

Thus all the integrals are bounded above by S, not only those with integer endpoints. They increase with R, again because $f \ge 0$. Their supremum is finite, and their limit as $R \to \infty$ equals that supremum: given a positive tolerance, choose an endpoint whose integral is within that tolerance of the supremum; every larger endpoint gives an integral between that value and the supremum. Hence the improper integral converges.

5. Explain divergence and derive the tail estimates.

If the integral tends to $+\infty$, the lower bound $S_N \ge \int_m^{N+1} f$ forces $S_N \to +\infty$. If instead $S_N \to +\infty$, the upper rectangle bound rearranges to $\int_m^N f \ge S_N - f(m)$, which forces the integral to $+\infty$ as well. Nonnegative increasing sums and integrals have no other type of divergence.

Finally, in the convergent case, apply the same unit-interval comparisons to the terms with indices $N+1, \dots, M$, where $M > N \ge m$:

$$\int_{N+1}^{M+1} f(x)\,dx \le \sum_{k=N+1}^{M} f(k) \le \int_{N}^{M} f(x)\,dx.$$

Letting $M \to \infty$ gives the stated remainder bounds. These limits are finite since the whole integral and series converge.

The same test applies if the assumptions hold only after some integer index: start at that index and use Exercise 14.9 to restore the finite initial segment.

Conclusion. For a continuous nonnegative nonincreasing function matching the terms, the series converges exactly when its improper integral is finite. The two neighboring tail integrals bound the omitted tail of the series.